Research Problem & Design Excellence

The problem

A vehicle count does not show how long a queue needs to clear.

Vehicle length and queue position change the required green duration.

Mixed Sri Lankan traffic needs local calibration, including tuk-tuks.

TrafficEye's design

Vehicle detection with a calibrated camera-view check and lane-region filtering.

An ordered queue with class- and position-specific discharge speeds.

FIFO clearance estimation with a data-derived safety margin.

An explainable recommendation and ambulance advisory for the officer.

Decision support: the officer makes the final decision. TrafficEye does not switch traffic signals.

Individual Research Component



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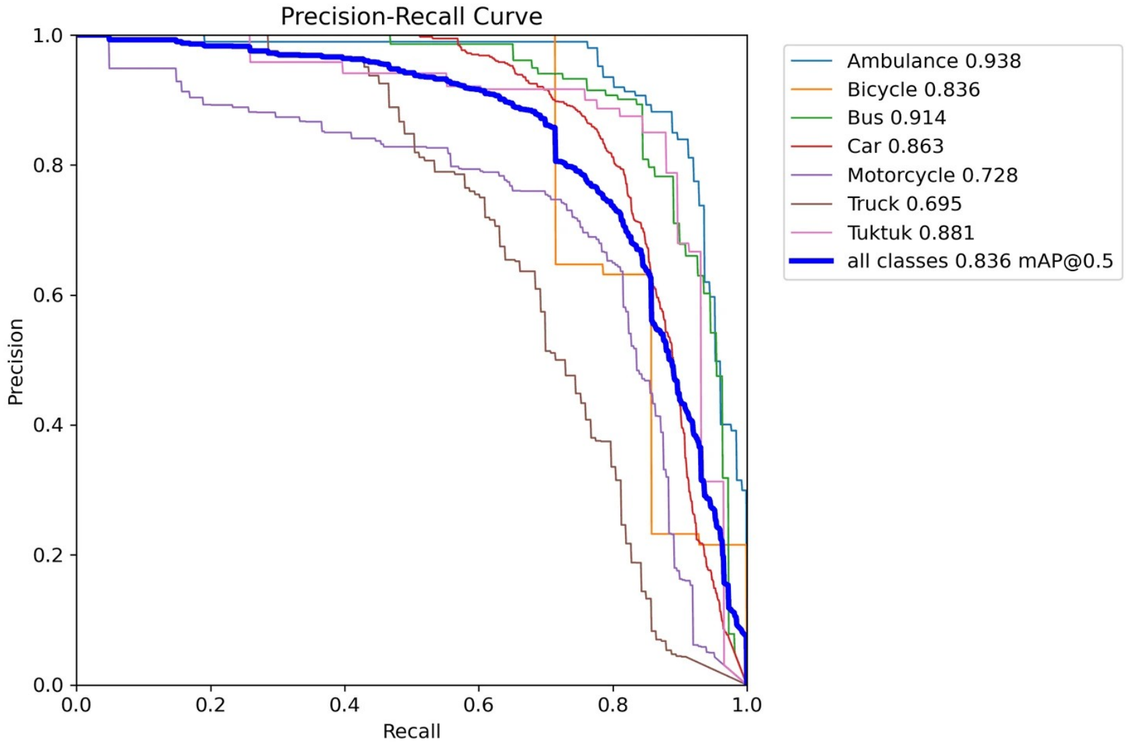
TrafficEye: Traffic Monitoring & Green Light Support System

Analyses recorded traffic video from a calibrated fixed camera to detect vehicles, order a single-lane queue, and estimate queue clearance time.

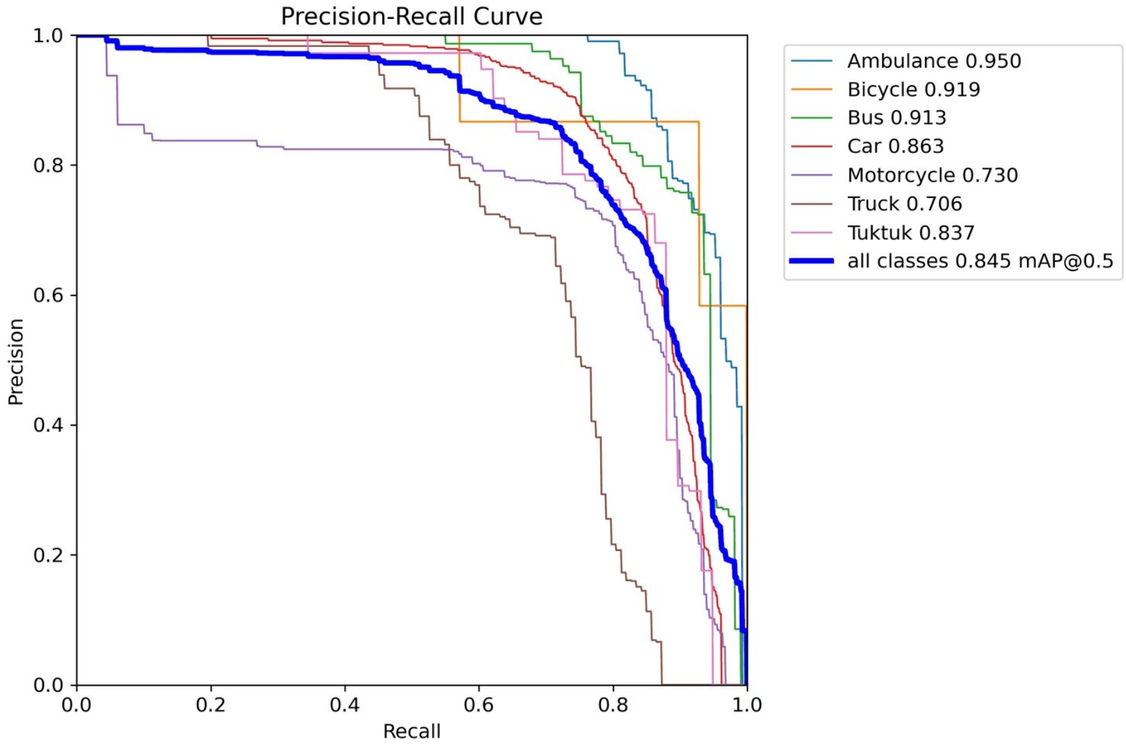
Recommends a green-light duration using locally calibrated discharge speeds, FIFO ordering and safety margin, with an ambulance priority advisory for the traffic officer.

YOLOv5n and YOLOv8n Benchmark

YOLOv5n baseline



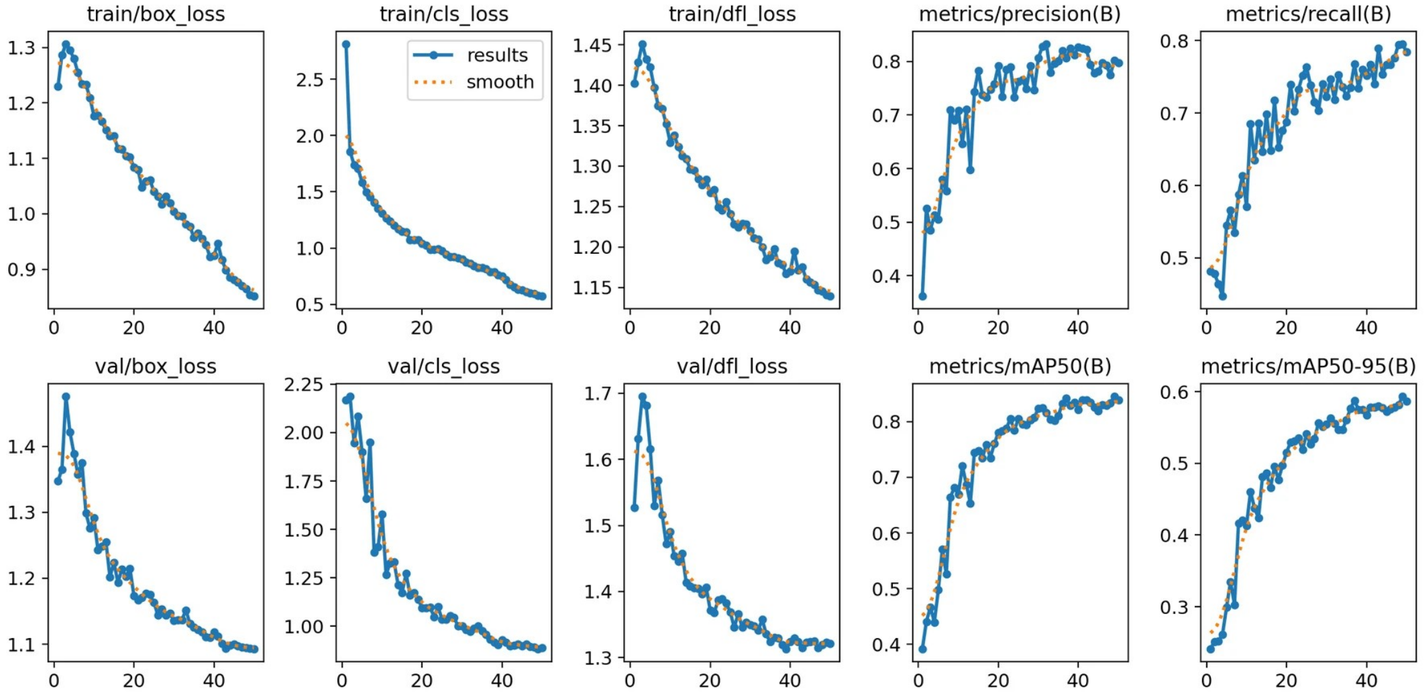
YOLOv8n selected model



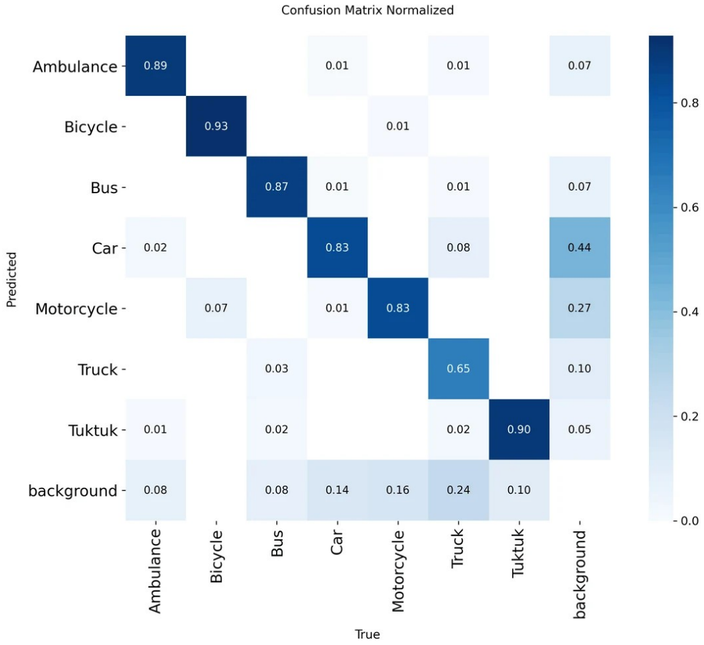
Earlier seven-class validation: mAP50 improved from 0.836 to 0.845 (+0.009).

YOLOv8n Training & Class-Level Evaluation

Training results

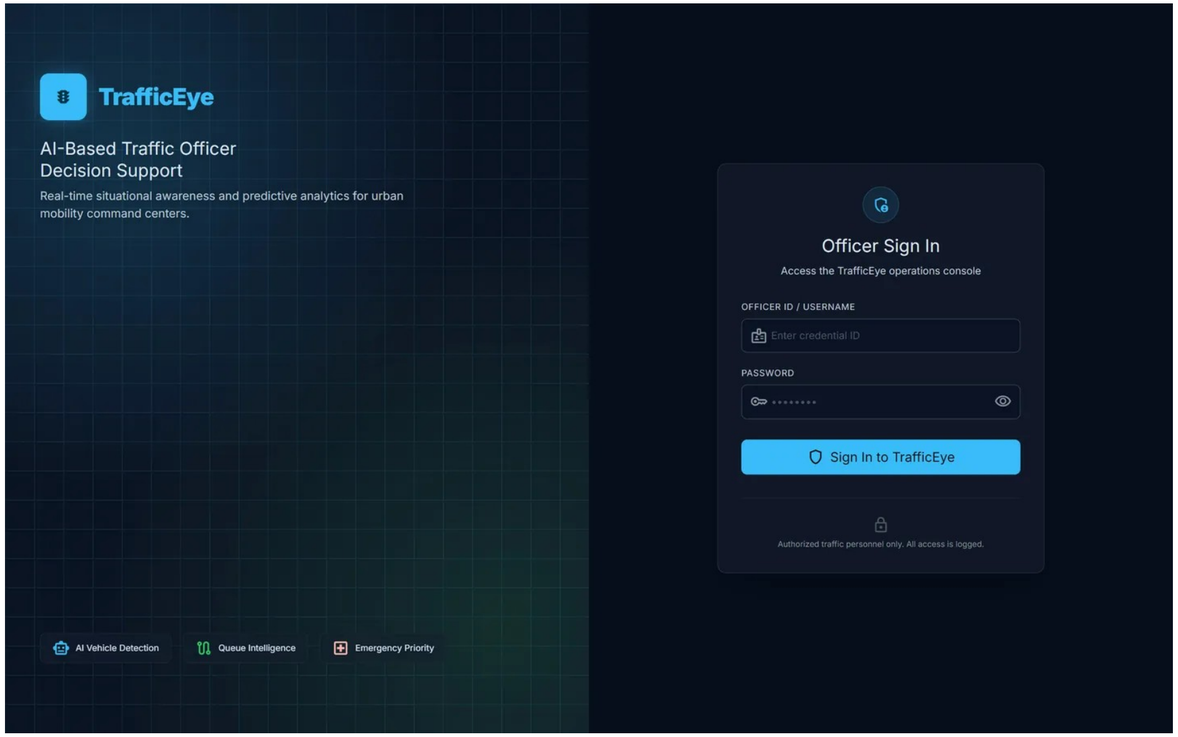


Normalized confusion matrix

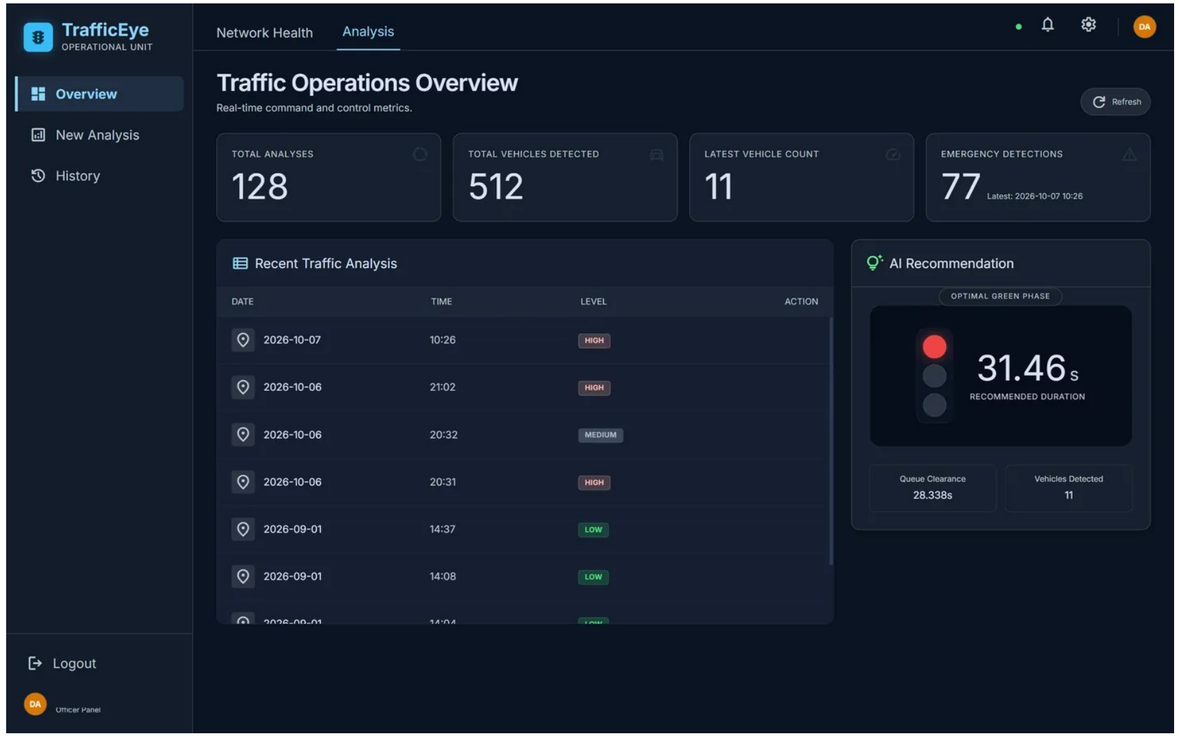


Implemented Officer Application

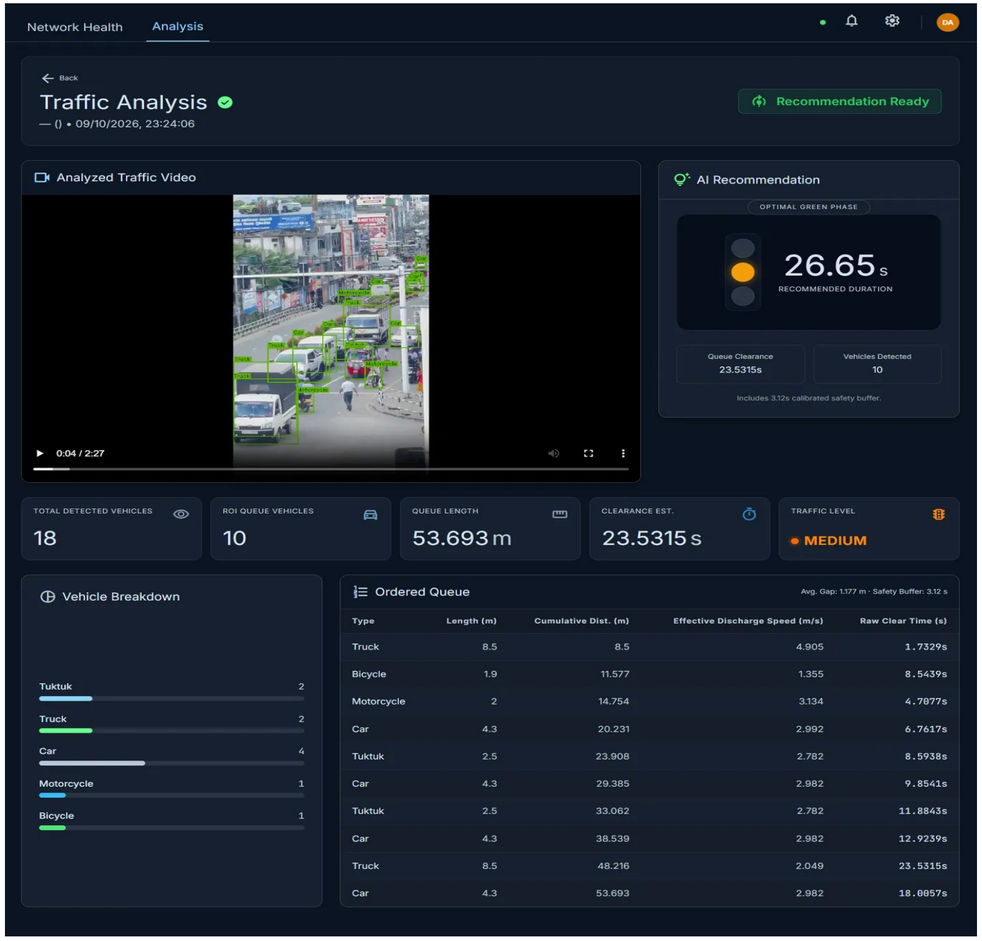
Officer sign-in

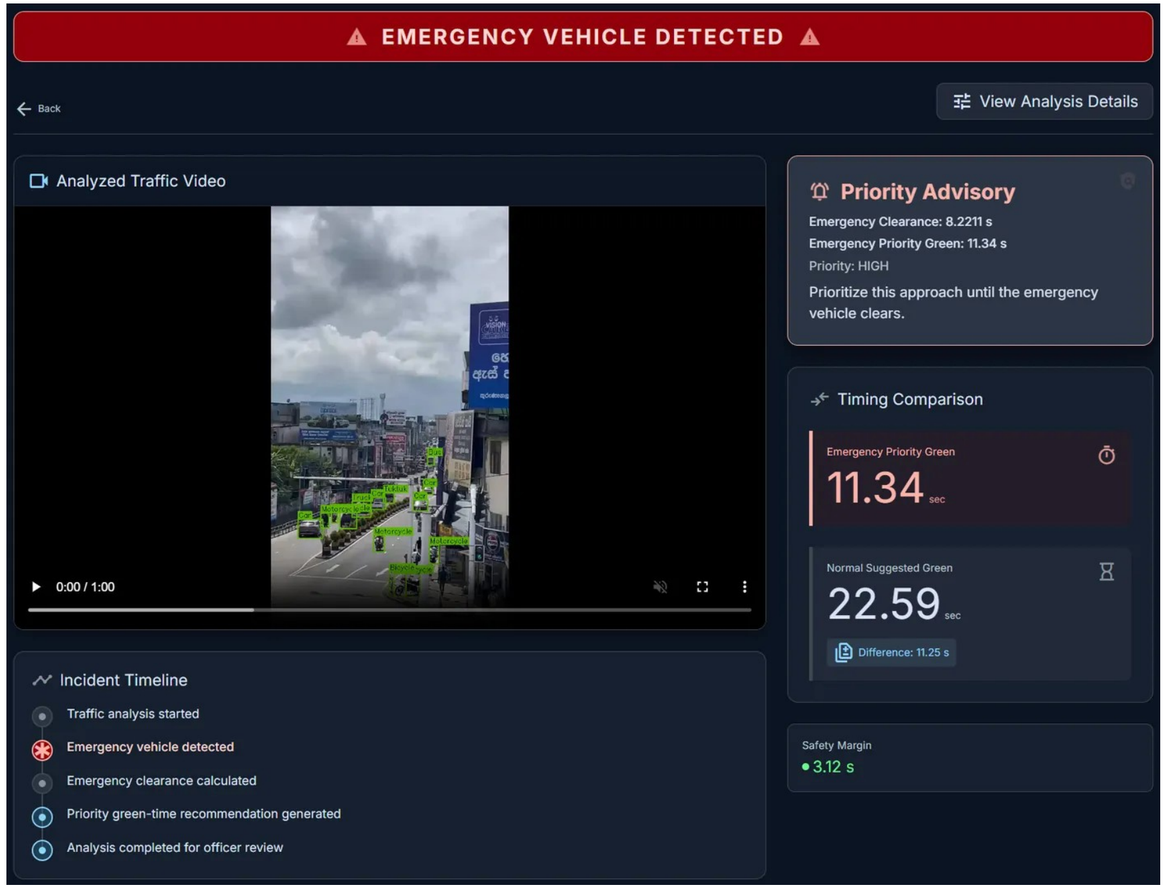


Analysis overview and history



Actual System Outputs





Results, Limitations & Future Improvements

Detector validation

Current October YOLOv8n checkpoint
Precision 0.7595 Recall 0.6788
mAP50 0.7477 mAP50–95 0.5148

Earlier seven-class development comparison
YOLOv8n mAP50 0.845
YOLOv5n mAP50 0.836

Limitations

Recorded playback and calibrated camera views. Single-lane scope. Detector metrics do not measure live-road impact.

Future improvements

Integrate the system with live CCTV feeds to provide real-time traffic analysis and green-light recommendations to traffic officers.